Fish | Candidate hypotheses

72 nodes / 495 edges

Upward: solid
Downward: dashed
Same scale: dotted
Micro
N19 Focal heading alignment with own escape direction
N12 Focal fish normalized component major-axis projection
N1 Contact-component size of focal fish
N66 Focal run-level influence percentile
N55 Focal angular velocity around school centroid
N48 Focal local clustering coefficient
N37 Focal heading deviation from school mean
N30 Focal distance to nearest frightened contact neighbor
N56 Focal signed tangential speed
N20 Focal absolute requested turn
N38 Focal heading alignment with contact neighbors
N2 Focal fish contact isolation flag
N21 Focal step heading change magnitude
N10 Focal fish normalized major-axis projection
N3 Focal component centroid to nearest shark distance
N28 Focal fish frightened indicator
N29 Focal remaining fright steps
N46 Focal contact degree
N39 Focal inverse distance to nearest shark
N11 Focal fish squared Mahalanobis distance
N47 Focal nearest contact-neighbor distance
N57 Focal radial velocity
N65 Focal run-level lagged influence score
N64 Focal lagged neighbor heading-change alignment
Meso
N41 Contact-component mean heading deviation
N23 Contact-component mean heading-escape alignment
N59 Contact-component rotational order
N13 Contact-component aspect ratio
N49 Contact-component edge density
N60 Contact-component mean angular momentum
N67 Contact-component mean lagged heading-change alignment
N50 Contact-component mean degree
N5 Contact-component size entropy
N6 Mean distance between component centroids
N24 Contact-component requested-turn variance
N42 Contact-component heading away-from-shark alignment
N14 Component major-axis escape alignment
N15 Contact-component major-axis length
N22 Contact-component escape-direction coherence
N32 Contact-component frightened-edge fraction
N33 Contact-component fright-centroid-distance correlation
N31 Contact-component fright prevalence
N40 Contact-component heading polarization
N51 Contact-component mean nearest-contact distance
N4 Mean contact-component size
N58 Contact-component mean angular velocity
N68 Contact-component sign-matched cascade fraction
N69 Contact-component influence-score variance
Macro
N72 Whole-school influence-score variance
N17 Whole-school anisotropy index
N18 Whole-school major-axis escape alignment
N27 Step change in school escape-direction coherence
N53 Whole-school global clustering coefficient
N25 Whole-school escape-direction coherence
N16 Whole-school aspect ratio
N34 Whole-school fright prevalence
N45 Step change in whole-school polarization
N8 Largest component fish fraction
N54 Whole-school mean nearest-contact distance
N43 Whole-school heading polarization
N52 Whole-school mean contact degree
N35 Whole-school frightened-edge fraction
N7 Number of contact components
N61 Whole-school mean angular velocity
N26 Heading-change magnitude vs shark proximity covariance
N36 Fright onset rate
N62 Whole-school rotational order
N63 Whole-school mean angular momentum
N71 Whole-school sign-matched cascade fraction
N9 Step change in contact-component count
N44 Run-level polarization vs mean shark proximity covariance
N70 Whole-school mean lagged heading-change alignment